

Shaz (Ethan Shahrooz) Moghaddam

Data Scientist · Python Developer

London, UK · 07519 109433 · shaz.moghaddam@gmail.com

linkedin.com/in/shazmoghaddam · github.com/ShazMoghaddam · shazmoghaddam.github.io

Right to work: British citizen · DBS check available on request

PROFILE

Data scientist and Python developer with a Data Science Bootcamp qualification from Imperial College London and HyperionDev (96% average). I build data projects end to end, from cleaning raw data and training models to deploying working, tested applications. Recent work includes VoltEdge and VoltScope, two energy data tools combining anomaly detection, AI-powered bill extraction and ESG reporting, and ClinIQ, a clinical trial monitoring dashboard. I have also published seven Android and Flutter apps on Google Play, so I am used to owning a product from first idea to release.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, Dart (Flutter), Kotlin, Java, HTML, CSS

Data and ML libraries: pandas, NumPy, SciPy, scikit-learn, TensorFlow, spaCy, NetworkX, Pydantic

Machine learning: linear and logistic regression, decision trees, random forests, gradient boosting, K-means clustering, anomaly detection (Isolation Forest), causal discovery (PC algorithm), NLP

AI tools: Claude, ChatGPT and Gemini, used daily in development and built into products through their APIs for structured data extraction and plain-English questions about data

Frameworks: FastAPI, Flask, Plotly Dash, Streamlit, React, Node.js, Electron, SQLAlchemy

Visualisation: Matplotlib, Seaborn, Plotly, Power BI, Tableau

Cloud and tools: Docker (multi-stage builds), AWS (ECS Fargate), GCP (Cloud Run), Render, Railway, Firebase, Git, Jupyter, REST APIs, automated testing

Mobile: Android (Kotlin, Java), Flutter, Google Play Store

SELECTED PROJECTS

VoltEdge, Energy Intelligence Platform

FastAPI · Plotly Dash · scikit-learn · NetworkX · SQLite · Docker · Claude API

- Monitors energy use across four industrial sites using 90 days of demo data, with anomaly detection for each site (Isolation Forest) and root-cause analysis through causal discovery (PC algorithm), scored by causal strength.
- ESG reporting covering Scope 2 emissions, GRI 302-1 and carbon intensity, structured for CSRD, plus an AI assistant (Claude) that answers plain-English questions about site data.
- JWT login with role-based access, an audit log of user actions and 903 passing tests. Deployed with Docker on Render.

github.com/ShazMoghaddam/volteedge · [Live demo](#)

VoltScope, Energy Bill Extraction and Validation

FastAPI · Claude API · Pydantic · SQLite · Docker

- Reads UK commercial energy bills from PDFs and phone photos (including iPhone HEIC) and extracts them into a fixed Pydantic schema using the Claude API. Also works as the bill module of VoltEdge.
- Rule-based validation checks MPANs, reconciles subtotals and VAT, and flags estimated readings, duplicate charges and contract renewal dates. A portfolio view highlights possible overcharges and compares suppliers.
- Review app with drag-and-drop upload, API-key authentication and JSON, Excel and CSV export. 50 passing tests based on real anonymised bills.

github.com/ShazMoghaddam/voltscope

ClinIQ, Clinical Trial Site Performance Dashboard

FastAPI · Plotly Dash · SQLAlchemy · scikit-learn · spaCy · Claude API · Docker

- Designed for mid-sized CROs in the UK and EU to compare trial sites on enrolment speed, risk and protocol deviations.
- Analyses deviation reports with spaCy, scores site risk with scikit-learn, and includes an AI assistant (Claude) for plain-English questions about trial data.

github.com/ShazMoghaddam/cliniq

SalaryAxis, UK Salary Benchmarking Platform

Python · Flask · React · pandas · NumPy · SciPy · Docker · ONS open data

- Salary benchmarking built on ONS ASHE data, with CPI inflation adjustment, regional and occupation breakdowns, gender pay gap trends and CAGR analysis.
- Python REST API with a React dashboard, packaged with multi-stage Docker builds for AWS ECS Fargate or GCP Cloud Run. No personal data is collected.

github.com/ShazMoghaddam/salaryaxis

CVLens, CV Parser and Analyser

Python · spaCy · pdfplumber · python-docx

- Reads CVs in PDF and Word format, extracts skills, detects section structure and flags weak spots, using spaCy entity recognition and rule-based matching.

github.com/ShazMoghaddam/CVLens

World Map, Interactive Choropleth Explorer

HTML · CSS · JavaScript · SVG

- Interactive map of 172 countries across seven data layers, including GDP per capita, HDI, CO₂ emissions and life expectancy, with filters, custom data import, shareable links and embedding. A single HTML file with no dependencies, MIT licensed.

github.com/ShazMoghaddam/world-map · [Live demo](#)

EXPERIENCE

Independent Product and Software Developer

2022 – Present

Remote · Google Play Store · Gumroad, Lemon Squeezy and Envato

- Built and published seven Android and Flutter apps on Google Play, owning each one from first idea through to release and updates.
- Sell digital products including finance tools (Market Pulse, TradeHours Pro), generative art (Typography Digits, KNOK Matrix) and screensavers (Aero Atlas), improving them based on customer feedback.
- Built the data and AI projects listed above during this period.

Career Break

Mar 2013 – 2022

Full-time primary carer for my family.

Sales Representative (Technology)

Mar 2011 – Mar 2013

John Lewis and Currys (via agency) · London

- Advised customers on technology products across several London stores, building communication and prioritisation skills.

EDUCATION AND CERTIFICATIONS

Data Science Bootcamp, Imperial College London and HyperionDev

June 2025

96% average. Covered SQL and databases, data cleaning and preprocessing, statistics, visualisation (Tableau and Python), supervised learning (linear and logistic regression, decision trees, random forests), K-means clustering and NLP, finishing with three capstone projects.

The Data Science Course: Complete Data Science Bootcamp (Udemy, 32 hours)

Oct 2025

iOS and Swift: The Complete iOS App Development Bootcamp (Udemy, 60.5 hours)

Jan 2024

100 Days of Code and The Complete Python Bootcamp (Udemy, 58 hours)

Oct 2022 – Jul 2023

Google Play Store Listing Certificate (Google Play Academy)

Apr 2026

ADDITIONAL

Writing: Seven personal essays on Medium about growth, identity, technology and perspective (medium.com/@shaz.moghaddam).

Availability: Full-time, freelance or contract. On-site, hybrid or remote in London, and open to international opportunities.